# ECO3121 final exam Cheat Sheet

1. Testing for group difference

- **Joint test with F-statistic**

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n-k-1)}$$

Chow Test:

$$F = \frac{[\text{SSR}_P - (\text{SSR}_1 + \text{SSR}_2)]}{\text{SSR}_1 + \text{SSR}_2} \cdot \frac{[n - 2(k+1)]}{k+1}$$

2. IV estimator:

$$\beta_1 = \frac{\text{cov}\,(Z, Y)}{\text{cov}\,(Z, X)}$$

Where Z denotes the IV and X denote the endogenous variable.

3. J-test:

1.Obtain the 2SLS estimator using all $m$ instruments; compute the predicted values $\hat{Y}_i$ using the actual $X$ 's
2.Compute the residual $\hat{u}_i = Y - \hat{Y}_i$
3.Regress $\hat{u}_i$ on $W_{1i}, \dots, W_{ri}, Z_{1i}, \dots, Z_{mi}$
4.Compute the $F$-statistic testing the hypothesis that the coefficients on $Z_{1i}, \dots, Z_{mi}$ are all zero
5.The $J$-statistic is $J = mF \sim \chi^2(m-p)$ , where p denotes # of endogenous variables.

4. Testing for heteroskedasticity

- **Steps in Computing the Breusch-Pagan Test**
  1. Estimate the equation $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + u$ by OLS, saving the OLS residuals, $\hat{u}_i$ .
  2. Regress $\hat{u}_i^2$ on all explanatory variables and compute the usual F test of joint significance of the explanatory variables.
  3. If the p-value of the test in step 2 is sufficiently small, reject the null of homoskedasticity and conclude Assumption MLR.5 fails.

5. FGLS

- **A Feasible GLS Procedure to Correct for Heteroskedasticity:**
  1) Estimate the model by OLS and obtain the residuals, $\hat{u}_i$.
  2) Create $\log(\hat{u}_i^2)$ by first squaring the OLS residuals and then taking the natural log.
  3) Run the regression of $\log(\hat{u}_i^2)$ on all explanatory variables and obtain the fitted values, $\hat{g}_i$
  4) Exponentiate the fitted values: $\hat{h}_i = exp(\hat{g}_i)$
  5) Find WLS estimation using weights $1/\hat{h}_i$